

Full-color natural light holographic video camera

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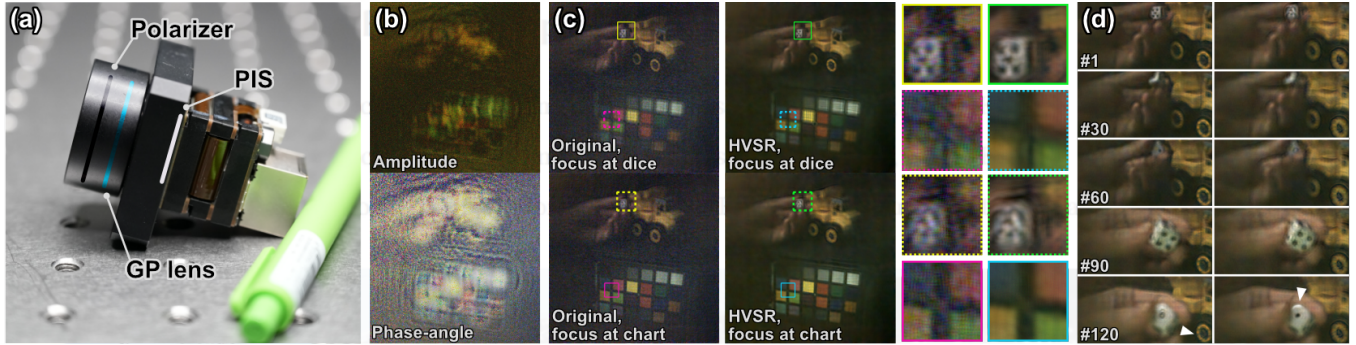


Figure 1: (a) Full-color natural light holographic video camera prototype. GP, geometric phase; PIS, polarized image sensor. (b) Acquired complex hologram. (c) Before and after the holographic video super-resolution (HVSR) process applied, with changing focus plane. (d) Demonstration of full-color video hologram acquisition, and its digital refocusing capability.

Abstract

We present a handheld holographic video camera that captures full-color, complex holographic video under ordinary lighting conditions. By adopting the self-interference incoherent holography principle, it eliminates the need for a laser light source, which is typically required for hologram recording, by integrating a geometric phase lens and a polarization image sensor, the system records complex-valued holograms in real time, without bias and twin image noises. Its centimeter-scale co-axial design enables truly mobile, handheld operation, while digital refocusing at arbitrary depths is achieved through numerical propagation with no prior knowledge. A video-based super-resolution algorithm further enhances the hologram's quality, preserving digital refocusing capabilities. The demonstration highlights the feasibility of everyday holographic content acquisition, bridging the gap between the laboratory and real-world applications.

CCS Concepts

• Hardware → Displays and imagers.

Keywords

Holographic video camera, Incoherent holography, Color holography

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1 Introduction

Hologram-based three-dimensional (3D) video has attracted significant attention as an ultimate media paradigm for capturing and delivering immersive visual experiences. Despite this promise, practical implementation remains challenging due to several technical hurdles. Among these challenges, a particularly critical issue is that the holographic content acquisition methods often require specialized lighting conditions and bulky equipment. These constraints make everyday use of holographic media difficult and hinder its broader adoption.

In developing a “holographic camera” suitable for everyday use, we identify three major requirements: (1) the ability to capture full-color video under natural (incoherent) lighting, (2) sufficient robustness to external vibrations so that the camera can be handheld, and (3) a compact design. Moreover, digital hologram demands higher spatial sampling compared to typical imaging systems, adding another layer of difficulty. Satisfying all these conditions simultaneously is essential for bringing holography closer to real-world applications.

One promising approach to meet these requirements is incoherent holography, which has been studied for decades [Kim et al. 2024; Rosen et al. 2021]. Rather than generating interference patterns by using a coherent reference beam and an object beam, incoherent holography relies on self-interference of incident light. This is

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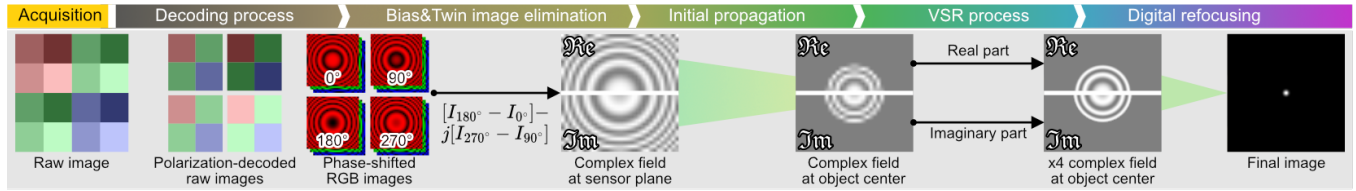


Figure 2: Frame processing pipeline

achieved by dividing and independently modulating portions of the incoming light before recombining them to form an interference pattern. Because the modulation takes place within the coherence length of the light source, even daylight or common indoor lighting can serve as illumination for holographic recording.

A key component for incoherent holography is wavefront modulation and phase shifting. Common approaches to wavefront modulation use conventional interferometers, spatial light modulators (SLMs), birefringent lenses, or liquid crystal lenses with polarization-selective properties. A dedicated phase-shifting module is essential to eliminate bias and twin-image noise, both of which arise inevitably from interference. Widely used phase-shifting elements include mirrors with actuators, SLMs, and variable retarders. Several research groups have demonstrated preliminary solutions that meet some of the requirements for incoherent hologram recording systems as a daily purpose holographic camera, but very few systems fully satisfy all the criteria described above.

In this paper, we present a holographic video camera capable of capturing real-time, full-color holographic video under natural lighting conditions. We also apply a holographic video super-resolution (VSR) algorithm that enhances the spatial resolution and image quality of the recorded complex holograms, producing high-quality digital holographic content suitable for next-generation holographic displays, yet digital refocusing at arbitrary depths is available through numerical propagation without any prior knowledge.

2 Acquisition System and processing pipeline

Our system integrates a geometric phase (GP) lens with a polarization image sensor to acquire self-interference patterns in real-time. The GP lens is a thin-film device whose focal power depends on the circular polarization state of the incident light. When linearly polarized light enters, half of the beam is effectively shaped as if by a concave lens, while the other half is shaped by a convex lens, thus enabling self-interference within a compact footprint [Choi et al. 2019].

By combining this GP lens with a polarized image sensor (PIS), we can record four phase-shifted images simultaneously. Each polarization channel encodes a different phase shift, allowing us to extract a complex hologram in real-time without bias or twin-image noise. In our demonstration, the optical system features a compact form factor, consisting solely of an input linear polarizer, a GP lens with a focal length of 0.5 m, and a color PIS (PHX-050S, Lucid Vision Labs). The total system length is about 1 cm. The acquisition speed is 25 fps. Figure 1(a) shows the prototype system.

Because the sensor arranges four polarization channels within each 2×2 pixel block, the effective pixel pitch doubles when extracting a single complex hologram. Consequently, the final complex hologram has half the resolution of the raw sensor image. Figure 1(b) shows the acquired complex hologram. To mitigate this reduction in spatial sampling rate, we apply a commercially available 2D VSR algorithm to the obtained hologram sequence [Labs 2025]. Hence, we first split the complex hologram into real and imaginary parts, feed each into the network separately, and then recombine them into the complex value. As a result, we obtain a four-times super-resolved complex hologram, where the digital refocusing is still available. Figure 2 describes the processing pipeline from the raw frame image. Figure 1(c) presents the comparison results before and after the VSR processing.

3 Demo Experience

We demonstrate the system in a standard video recording environment under typical lighting conditions, showcasing its capability to record real-time holographic video of arbitrary objects. Visitors can grab the camera and capture holographic videos of their surroundings and observe intermediate processing steps as well as the final output through a graphical user interface. This demonstration setup allows visitors to experiment with digital refocusing, highlighting the practical capabilities of our handheld incoherent holographic camera system. Through this hands-on demo, participants can glimpse a future where compact, handheld holographic cameras capture immersive 3D scenes under natural light, paving the way for next-generation hologram media services.

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